

AI Content Creator Application

SUBMITTED BY	ENVIRONMENT	HOST PORT	VERIFICATION
Mohammed Kaderji	Jupyter • Streamlit • Gemini	localhost:8502	✓ Verified & Tested

1. Objective & Folder Structure

Objective: Build a web application that automatically generates copy for LinkedIn Posts, Instagram Captions, Twitter Posts, Professional Emails, and Blog Outlines using Streamlit and the Gemini API.

Folder Structure: AI_Content_Generator/ → app.py, requirements.txt, README.md

2. Step-by-Step Implementation Guide

Step 1: Install Required Libraries

Open Jupyter Notebook, click inside an empty cell, and install the required modules:

```
!pip install streamlit google-generativeai
```

Press **Shift + Enter**. Wait until the `[*]` changes to a fixed number like `[1]`.

Step 2: Create Folder & Files

Inside Jupyter, set up the project folder and requirements file:

- Run `import os; os.makedirs("AI_Content_Generator", exist_ok=True)`
- Run `%%writefile AI_Content_Generator/requirements.txt` containing streamlit and google-generativeai.

Step 3: Write app.py (Base Code + Tone Challenge)

In a new cell, write the code using `%%writefile AI_Content_Generator/app.py`.

Solving the Project Challenge: Added an extra `st.selectbox()` for **Tone** (*Professional, Funny, Formal, Friendly*) to allow full control over the writing persona.

Step 4: Run Application (Server Launch)

Open Command Prompt (cmd), navigate to the directory, and start the Streamlit server on port 8502:

```
cd C:/Users/Hp/AI_Content_Generator
```

```
streamlit run app.py --server.port 8502
```

Step 5: Verify Live Output

Open your web browser at `http://localhost:8502`, type topic *"World evolving with ai"*, choose *"LinkedIn Post"*, select tone *"Professional"*, and click **Generate**.

3. Source Code Implementation (`app.py`)

```
import streamlit as st
import google.generativeai as genai

# 1. Configure Gemini API Credentials
genai.configure(api_key="YOUR_API_KEY")

# 2. Initialize Gemini 3.6 Flash model
model = genai.GenerativeModel("gemini-3.6-flash")

# 3. Application User Interface
st.title("👋 AI Content Generator")
st.write("Generate high-quality copy for different platforms with a single click.")

topic = st.text_input("Enter Topic", placeholder="e.g., World evolving with AI")

content_type = st.selectbox(
    "Content Type",
    [
        "LinkedIn Post",
        "Instagram Caption",
        "Twitter Post",
        "Professional Email",
        "Blog Outline"
    ]
)

# Project Challenge: Tone Selector
tone = st.selectbox(
    "Select Tone",
    ["Professional", "Funny", "Formal", "Friendly"]
)

# 4. Generate Content on Button Click
if st.button("Generate"):
    if topic:
        prompt = f"""
        You are a professional content writer.
        Create a {content_type} in a {tone} tone.

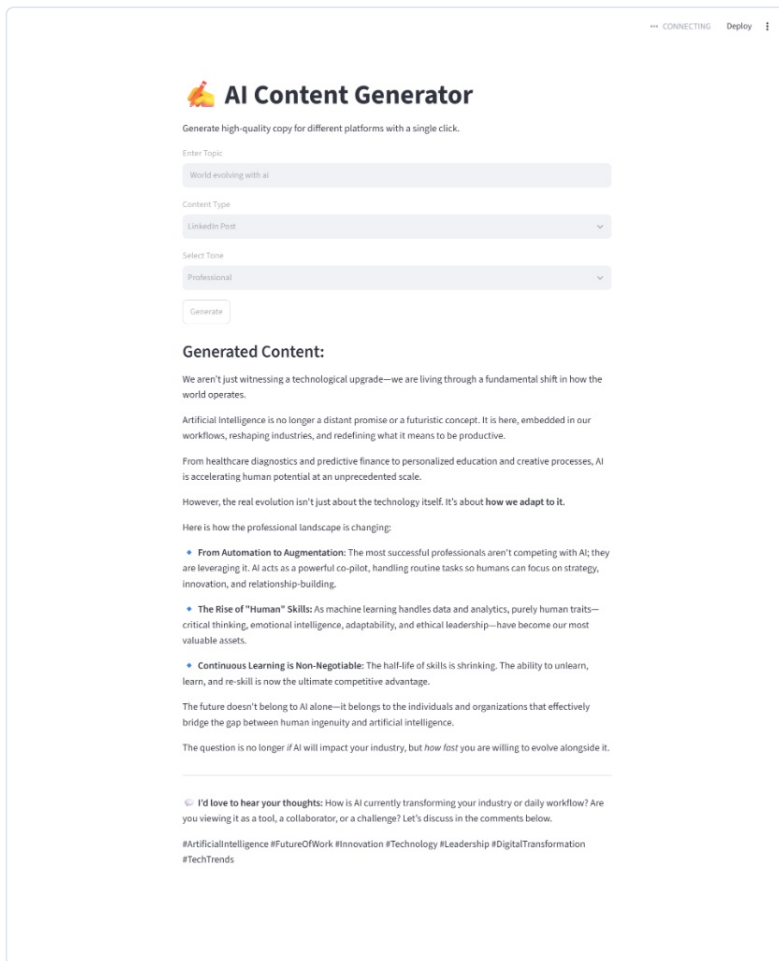
        Topic:
        {topic}
        """

        with st.spinner("Generating content..."):
            response = model.generate_content(prompt)
            st.markdown("### Generated Content:")
            st.write(response.text)
    else:
        st.warning("Please enter a topic first.")
```

4. Live Execution Proof & Verification (Port 8502)

Evaluation & Test Parameters:

- **Tested Topic:** *"World evolving with ai"*
- **Selected Platform & Format:** *LinkedIn Post*
- **Challenge Tone Selected:** *Professional*
- **Observed Output & Quality:** Generated an executive-grade LinkedIn post covering workforce augmentation, the rise of human skills, non-negotiable continuous learning, structured engagement calls-to-action, and relevant professional hashtags (#ArtificialIntelligence #FutureOfWork #DigitalTransformation).



The screenshot shows a web application titled "AI Content Generator" with a sub-header "Generate high-quality copy for different platforms with a single click." The interface includes three input fields: "Enter Topic" (containing "World evolving with ai"), "Content Type" (set to "LinkedIn Post"), and "Select Tone" (set to "Professional"). A "Generate" button is located below these fields. The "Generated Content:" section displays a professional LinkedIn post draft. The draft begins with a paragraph about technological shifts, followed by a paragraph on AI's integration into workflows. It then lists three key points: "From Automation to Augmentation," "The Rise of 'Human' Skills," and "Continuous Learning is Non-Negotiable." The draft concludes with a call to action and a list of relevant hashtags: #ArtificialIntelligence, #FutureOfWork, #Innovation, #Technology, #Leadership, #DigitalTransformation, and #TechTrends.

CONNECTING Deploy

AI Content Generator

Generate high-quality copy for different platforms with a single click.

Enter Topic

World evolving with ai

Content Type

LinkedIn Post

Select Tone

Professional

Generate

Generated Content:

We aren't just witnessing a technological upgrade—we are living through a fundamental shift in how the world operates.

Artificial Intelligence is no longer a distant promise or a futuristic concept. It is here, embedded in our workflows, reshaping industries, and redefining what it means to be productive.

From healthcare diagnostics and predictive finance to personalized education and creative processes, AI is accelerating human potential at an unprecedented scale.

However, the real evolution isn't just about the technology itself. It's about **how we adapt to it**.

Here is how the professional landscape is changing:

- **From Automation to Augmentation:** The most successful professionals aren't competing with AI; they are leveraging it. AI acts as a powerful co-pilot, handling routine tasks so humans can focus on strategy, innovation, and relationship-building.
- **The Rise of "Human" Skills:** As machine learning handles data and analytics, purely human traits—critical thinking, emotional intelligence, adaptability, and ethical leadership—have become our most valuable assets.
- **Continuous Learning is Non-Negotiable:** The half-life of skills is shrinking. The ability to unlearn, learn, and re-skill is now the ultimate competitive advantage.

The future doesn't belong to AI alone—it belongs to the individuals and organizations that effectively bridge the gap between human ingenuity and artificial intelligence.

The question is no longer *if* AI will impact your industry, but *how fast* you are willing to evolve alongside it.

🗣️ **I'd love to hear your thoughts:** How is AI currently transforming your industry or daily workflow? Are you viewing it as a tool, a collaborator, or a challenge? Let's discuss in the comments below.

#ArtificialIntelligence #FutureOfWork #Innovation #Technology #Leadership #DigitalTransformation #TechTrends

Figure 1.0: Live web application execution proof verifying custom topic input, tone selection challenge, and structured LinkedIn post generation.